Ch 3.3: Even More Linear Regression

Lecture 7 - CMSE 381

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Announcements

Last time:

- 3.2 Multiple Linear Regression

3.2.2

Announcements:

- Third homework posted on github
- Office hours

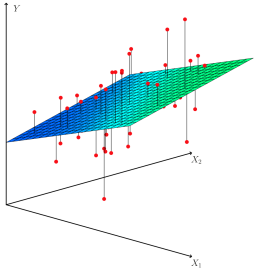
Covered in this lecture

- Qualitative predictors
- Extending the linear model with interaction terms
- Hierarchy principle
- Polynomial regression

Section 1

Review from last time

Linear Regression with Multiple Variables



- Predict Y on a multiple variables X

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p x_p + \varepsilon$$

- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1, \cdots$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_p x_p$

- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$
- RSS is minimized at *least squares coefficient estimates*

Questions to ask of your model

Don't care about β_0

$H_0: \beta_1 = \beta_2 = \cdots = \beta_p = 0$

H_a: at least one $\neq 0$

1. Is at least one of the predictors $X_1, \cdots, X_p$ useful in predicting the response? — F-statistic, p-values
2. Do all the predictors help to explain Y, or is only a subset of the predictors useful? — Subset selection
3. How well does the model fit the data? — RSE, R^2
4. Given a set of predictor values, what response value should we predict, and how accurate is our prediction? — Confidence interval vs prediction interval

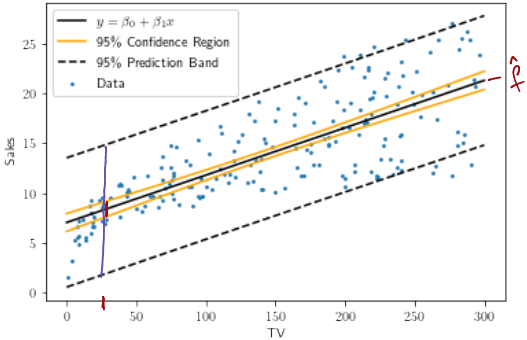
Confidence vs Prediction Model

Confidence Interval

The range likely to contain the population parameter (mean, standard deviation) of interest.

Prediction Interval

The range that likely contains the value of the dependent variable for a single new observation given specific values of the independent variables.



Specific to the Advertising Data

Confidence interval: quantify the uncertainty surrounding the average sales over a large number of cities.

Advertising example:
If \$100K is spent on TV, and \$20K on radio, **in each of n cities**

95% CI for sales:
[10,985, 11,528].

Prediction Interval: quantify the uncertainty in sales for a particular city.

Advertising example:
Given that \$100,000 is spent on TV advertising and \$20,000 is spent on radio advertising in **Gotham City**

95% prediction interval for Gotham:
[7,930, 14,580].

Section 2

Qualitative Predictors

Reminder: Qualitative vs Quantitative predictors

Quantitative:

Takes values in $\mathbb{R}$, $\mathbb{Z}$
real #'s, integers

- mpg
- weight
- Age
- year
- temp

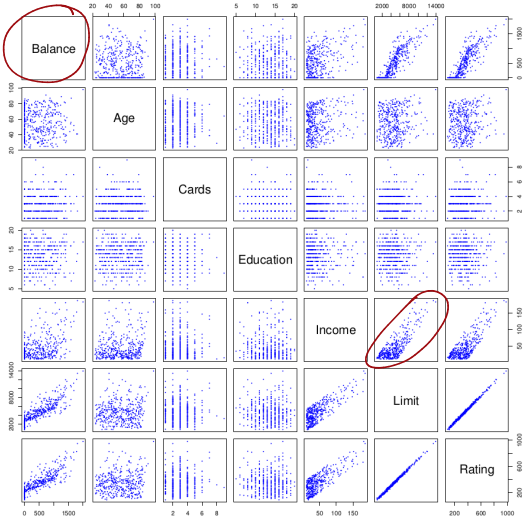
Order matters!

Qualitative/Categorical:

Set where order doesn't matter

- Anything yes|no true|false
- Countries
- Brand names
-

New data set! Credit card balance



- ~~`own`: house ownership~~
- `student`: student status
- `status`: marital status
- ~~`region`: East, West, or South~~

What if....

... your input variables aren't quantitative?

- Home ownership
- Student status
- Major
- Gender
- Ethnicity
- Country of origin

Example

Investigate differences in credit card balance between people who own a house and those who don't, ignoring the other variables.

One-hot encoding

Create a new variable

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

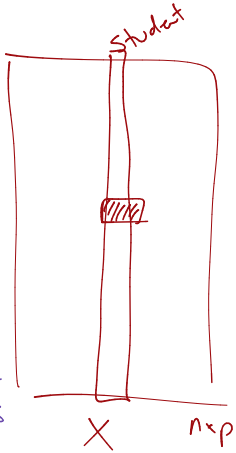
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases}$$

$\beta_0 + \beta_1$ is the average balance among students

β_0 is avg. balance among non students

β_1 is the diff in balance bt the groups



Interpretation

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	480.3694	23.434	20.499	0.000	434.300	526.439
Student[T.Yes]	396.4556	74.104	5.350	0.000	250.771	542.140

Model:

$$y = 480.36 + 396.46 \cdot x_{student}$$

Avg Balance for students is

\$480 + \$396 = \$876

Avg Balance for non-students is \$480

Who cares about 0/1?

Old version: 0/1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Alternative version: ± 1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ -1 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 - \beta_1 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Qualitiative Predictor with More than Two Levels

Region:

	x_{i1} – South	x_{i2} – West	x_{i3}
South	1	0	0
West	0	1	0
East	0	0	1

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

$$= \begin{cases} \beta_0 + \beta_1 \cancel{x_{i1}} + \varepsilon_i & \text{if } i\text{th person from South} \\ \beta_0 + \beta_2 \cancel{x_{i2}} + \varepsilon_i & \text{if } i\text{th person from West} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person from East} \end{cases}$$

More on multiple levels

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	531.00	46.32	11.464	< 0.0001
`region[South]`	-18.69	65.02	-0.287	0.7740
`region[West]`	-12.50	56.68	-0.221	0.8260

- Always want # less variable than # of levels
- Level with no variable is called the baseline

Baseline: East

avg balance of East

\$531

Avg for South

\$531 − 18.69 = \$513

High p-values here

No stat evidence of diff by region

Do code section on "Playing with multi-level variables"

Section 3

Extending the linear model

Assumptions so far

Back to our Advertising data set

$$\hat{Y}_{sales} = \beta_0 + \beta_1 \cdot X_{TV} + \beta_2 \cdot X_{radio} + \beta_3 \cdot X_{newspaper}$$

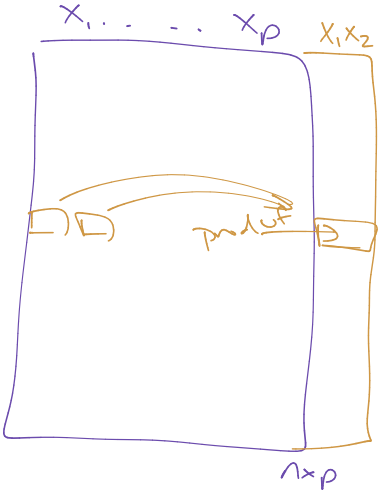
Assumed (implicitly) that the effect on sales by increasing one medium is independent of the others.

What if spending money on radio advertising increases the effectiveness of TV advertising? How do we model it?

Interaction Term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$



Interaction term

R^2 values

w/ interaction term 96.8%

w/o interaction term 89.7%

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	6.7502	0.248	27.23	< 0.0001
`TV`	0.0191	0.002	12.70	< 0.0001
`radio`	0.0289	0.009	3.24	0.0014
`TV×radio`	0.0011	0.000	20.73	< 0.0001

Low p-value for interaction term

Strong evidence that $\beta_3 \neq 0$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

Interpretation

	Coefficient	Std. error	t-statistic	p-value
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`TV`	0.0191	0.002	12.70	< 0.0001
`radio`	0.0289	0.009	3.24	0.0014
`TV`$\times$`radio`	0.0011	0.000	20.73	< 0.0001

Do the section on “Interaction Terms”

Hierarchy principle

Sometimes p-value for interaction term is very small, but associated main effects are not.

The hierarchy principle:

If we include the interaction term we should include main effects no matter what.

Interaction term for qualitative variables

Without interaction term

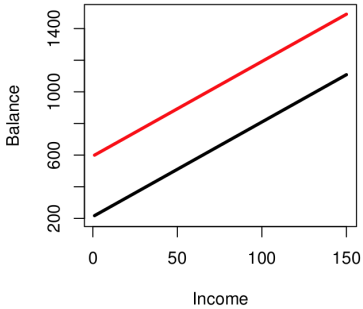
For credit data set:

Predict balance using income (quantitative) and student (qualitative)

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_2 & \text{if student} \\ 0 & \text{if not} \end{cases}$$

$$\approx \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_0 + \beta_2 & \text{if student} \\ \beta_0 & \text{if not} \end{cases}$$

β_1 Income $+ (\beta_0 + \beta_2)$



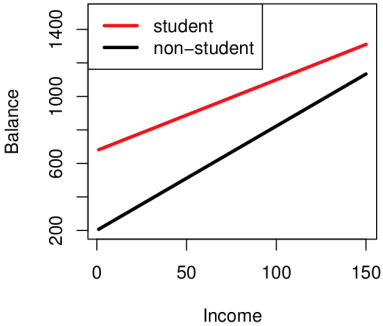
Interaction term for qualitative variables

With interaction term

For credit data set:
Predict balance using income (quantitative) and student (qualitative)

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_2 + \beta_3 \cdot \texttt{income}_i & \text{if student} \\ 0 & \text{if not} \end{cases}$$

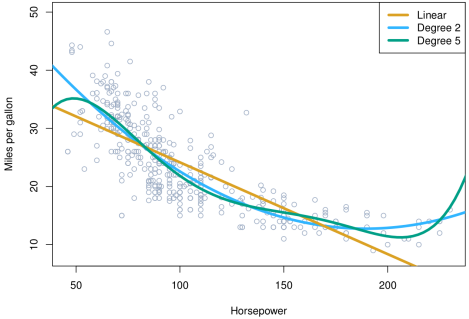
$$\approx \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot \texttt{income}_i & \text{if student} \\ \beta_0 + \beta_1 \cdot \texttt{income}_i & \text{if not} \end{cases}$$



Nonlinear relationships

More in Ch 7

$$\texttt{mpg} = \beta_0 + \beta_1 \cdot \texttt{horsepower} + \beta_2 \cdot \texttt{horsepower}^2 + \varepsilon$$



	Coefficient	Std. error	t-statistic	p-value
`Intercept`	56.9001	1.8004	31.6	< 0.0001
`horsepower`	-0.4662	0.0311	-15.0	< 0.0001
`horsepower`^{2}	0.0012	0.0001	10.1	< 0.0001

Next time

Lec #	Date		Topic	Reading	Homeworks
1	W	Aug 31	Intro / First day stuff / Python Review Pt 1	1	
2	F	Sep 2	What is statistical learning?	2.1	
	M	Sep 5	No class - Labor day		
3	W	Sep 7	Assessing Model Accuracy	2.2.1, 2.2.2	HW #1 Due
4	F	Sep 9	Linear Regression	3.1	
5	M	Sep 12	More Linear Regression	3.1/3.2	
6	W	Sep 14	Even more linear regression	3.2.2	HW #2 Due
7	F	Sep 16	Probably more linear regression	3.3	
8	M	Sep 19	Intro to classification, Logisitic Regression	2.2.3, 4.1, 4.2, 4.3	
9	W	Sep 21	More logistic regression		HW #3 Due
10	F	Sep 23	Review		
11	M	Sep 26	***Midterm #1***		
12	W	Sep 28	[No class, Dr Munch out of town]		
13	F	Sep 30	[No class, Dr Munch out of town]		
14	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
15	W	Oct 5	k-fold CV	5.1.3	
16	F	Oct 7	More k-fold CV	5.1.4	
17	M	Oct 10	CV for classification	5.1.5	HW #4 Due
18	W	Oct 12	Resampling methods: Bootstrap	5.2	
19	F	Oct 14	Subset selection	6.1	
20	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
21	W	Oct 19	Shrinkage: Lasso	6.2.2	
22	F	Oct 21	Dimension Reduction	6.3	